Verified 7895728366

Roll No. | | | | | | | |

M. Tech MID EXAMINATION .....-2025

Name of the Course: M.Tech
Name of the Paper: Composite Materials
Time: 1.5 Hours

Semester: III
Paper Code: OEME 04
Maximum Marks: 50

Note:-

- All questions are compulsory.
- Answer any one sub questions among a, & b in each main question.
- Each question carries 10 marks.

Q1	**10 marks**	**CO1**
(a)	Explain the fabrication technologies used for polymer matrix composites.	
	OR	
(b)	Discuss key applications of composites in biomedical and marine industries.	
Q2	**10 marks**	
(a)	Compare ceramic and metal matrix composites with respect to strength and thermal resistance.	**CO 2**
	OR	
(b)	Evaluate the mechanical behavior of a typical fiber-matrix interface.	
Q3	**10 Marks**	
(a)	Discuss strength prediction models for fiber-reinforced composites.	**CO 1/2**
	OR	
(b)	Illustrate the in-plane and out-of-plane stress states in laminated composites.	
Q4	**10 marks**	
(a)	Describe microstructural evolution in composites under static and dynamic loading.	**CO 3**
	OR	
(b)	Discuss methods for assessing impact resistance of laminates.	
Q5	**10 marks**	
(a)	Perform reliability analysis of a composite beam under variable loading.	**CO 2/3**
	OR	
(b)	Explain the significance of boundary conditions in the vibration analysis of plates.	